CONCEPT PAPER

Title: Predicting Course Persistence of Graduate Students from Learning Management System Activity Using Machine Learning

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Background and Rationale

Some graduate students stop attending an online or blended course before the term ends, and the program learns about it only after grades are filed. Activity logs from the learning management system can be used to train machine learning classifiers that flag students who are at risk of dropping a course. Educational data mining has shown that simple features such as login frequency, assignment submission timing, and forum posts carry useful signals.

Statement of the Problem

The problem is that the Graduate School has no early, data-based signal of which students are likely to stop participating. The study asks how accurately logistic regression and random forest models can predict course persistence.

Objectives

1. Build a dataset of learning management system activity features for graduate courses. 2. Compare the accuracy of logistic regression and random forest classifiers in predicting persistence. 3. Identify the activity features that contribute most to the prediction.

Methodology

This is a quantitative predictive research design. Data collection uses anonymized activity logs and final course status from past terms. The data will be split into training and test sets, features will be selected, and models will be compared using accuracy, precision, recall, and cross-validation. The analysis follows the data privacy rules of the university.

Significance and Scope

An early signal helps instructors and program coordinators reach out to students in time. The scope covers graduate courses of one college and does not include undergraduate data.

References

Romero, C., and Ventura, S. (2020). Educational data mining and learning analytics: an updated survey. Baker, R. (2019). Challenges for the future of educational data mining. Breiman, L. (2001). Random forests.